



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
20-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **27** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

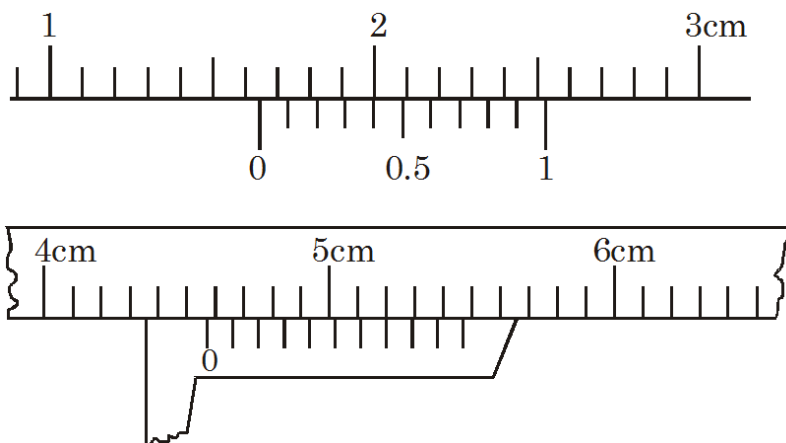
BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

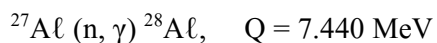
1. A student measured the length and width of a rectangular plate using a vernier callipers whose readings are shown below. What is the area of the plate?



- (A) 7.82 cm^2 (B) 7.21 cm^2 (C) 7.51 cm^2 (D) 7.49 cm^2
2. Two parallel plate capacitors filled with air with the same capacity C are connected in series and connected to a voltage source. Then space between the plates of one of the capacitors is filled with dielectric with a dielectric constant $k = 9$. How many times does the electric field E in that capacitor change?
- (A) Becomes $1/5$ times (B) Becomes 5 times
 (C) Becomes 9 times (D) Becomes $1/9$ times

Space for Rough Work

3. The rest mass of ${}_{13}^{27}\text{Al}$ is 26.98154 amu. Find the mass of the product nuclei for the following reactions :



The mass of neutron is 1.00866 amu. Take 1 amu = 930 MeV

- (A) 27.99820 amu (B) 27.99660 amu
(C) 27.98220 amu (D) 28.13480 amu
4. High-quality camera lenses are often coated to prevent reflection. A lens has an optical index of refraction of 1.72 and a coating with an optical index of refraction of 1.31. For near-normal incidence, the minimum thickness of the coating to prevent reflection for wavelength of 5.3×10^{-7} m is
- (A) 0.75 μm (B) 0.2 μm (C) 0.1 μm (D) 1.75 μm
5. A sound wave travels in HCl gas contained in a vessel at a speed of 300 m/s. When the temperature is doubled, half of the molecules dissociate. What is the speed of sound in the mixture? Molar mass of H = 1 gm, Molar mass of Cl = 35.5 gm.

- (A) $300\sqrt{\frac{15}{7}}$ (B) $300\sqrt{\frac{60}{7}}$ (C) $300\sqrt{\frac{255}{77}}$ (D) $300\sqrt{\frac{185}{77}}$

Space for Rough Work

6. Two closed organ pipes A and B have lengths 75 cm and 175 cm respectively. The 3rd harmonic of A corresponds to nth overtone of B. Find the value of n.
- (A) $n = 3$ (B) $n = 4$ (C) $n = 5$ (D) $n = 2$
7. The left end of a uniform long string of mass per unit length $\mu = 10^{-2}$ kg/m connected to a vibrating source. The vibrating source produces sinusoidal transverse wave of amplitude $A = 2$ cm on the string. The tension in the string is 100 N and frequency of vibrating source is 500 Hz. Neglecting all dissipative factors, the energy supplied by vibrating source to the string in 10 sec after it starts vibrating is (take $\pi^2 = 10$)
- (A) 2×10^4 J (B) 2×10^3 J (C) 200 J (D) 20 J

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

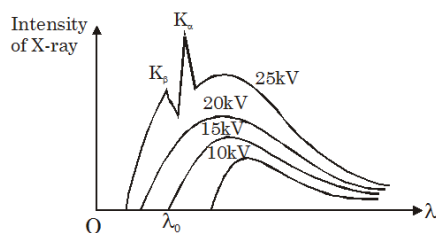
1. In a biprism experiment the micrometer readings for zero order fringe and tenth order fringe are 2.37 mm and 3.55 mm respectively, when light of $\lambda = 5.9 \times 10^{-5}$ cm is used. if λ is changed to 7.5×10^{-5} cm,
 - (A) The central maxima shifts to 2.22 mm
 - (B) The tenth order maxima shifts to 3.87 mm
 - (C) The central maxima remains at 2.37 mm
 - (D) The tenth order maxima can be again obtained at 3.55 mm if we use another biprism of same refractive index, but smaller angle.

Space for Rough Work

2. An air filled capacitor plate has a plate separation of 0.885 ± 0.009 mm. The area of the plates is 0.040 ± 0.001 m². It is connected to a voltage source of voltage 10.0 ± 0.1 V. Choose the correct statement(s)
- (A) The charge has a relative error of 4.5% approx
(B) The energy stored has a relative error of approx. 5.5%
(C) The force between the plates has a relative error of approx.. 5.5%
(D) If we insert a dielectric of dielectric constant $k = 4$ which fills the region between them completely, the random error in energy increases.
3. Young double slit apparatuses are used to measure the thickness of a thin glass plate of refractive index 1.5 (Both refractive index of plate and that of air are known precisely. Refractive index of air = 1). The central maxima shifts by a distance y as seen in the table below.
- (a) $D = 1.0$ m, $d = 2.50$ mm, $y = 1.0$ mm
(b) $D = 1.2$ m, $d = 3.00$ mm, $y = 1.1$ mm
(c) $D = 1.1$ m, $d = 1.65$ mm, $y = 1.7$ mm
(d) $D = 2.0$ m, $d = 4.00$ mm, $y = 1.2$ mm
- From this data, comment on the thickness of the plate.
- (A) $t = 5.1$ μ m.
(B) The value of t has 2 significant digits
(C) The random error in t has 1 significant digit
(D) The random error is 0.2 μ m

Space for Rough Work

4. X-ray spectrum obtained from a metallic target has been shown in figure. The spectrum has been obtained for four different accelerating voltages in the X-ray tube : 25 kV, 20 kV, 15 kV and 10 kV.



- (A) The binding energy of electron in K shell is $25 \text{ keV} \geq E > 20 \text{ keV}$
 (B) The binding energy of electron in K shell is $25 \text{ keV} > E > 20 \text{ keV}$
 (C) The Value of λ_0 is approx. 0.83 nm
 (D) The value of minimum wavelength when the voltage applied across the x ray tube is 10kV is $3\lambda_0/2$.
5. A charged particle of mass 10 gm and charge +10 C is projected inside the smooth surface of a hollow long cylindrical shell carrying a current per unit length of 1000 A/m. The current in the shell is along the circumference as shown. The shell has a radius of 5 m. Neglect gravity.



- (A) The particle will cross the axis of shell if its velocity is 1.57 m/s in the direction shown in the figure.
 (B) The minimum initial velocity of the charge particle so that the charge particle may always remain in contact with the shell is 6.28 m/s
 (C) If the charge is projected in the direction opposite to the shown direction, it will always remain in contact with the shell
 (D) The particle will cross the axis of shell if its velocity is 3.14 m/s in the direction opposite to that shown in the figure.

Space for Rough Work

6. A student named Amit found a convex lens lying somewhere and decided to find the focal length of it using optical bench. In step 1, to get rough estimate of focal length, he focused inverted image of a distant window on a screen and found distance between lens and screen to be somewhere in between 15 cm and 20 cm. Now, in step 2 he placed the object needle at a distance of 35 cm from the lens. In step 3, he looked at image of object from the end of the optical bench from side opposite to that of object and placed image pin such that tip of image and image pin coincided. After step 3, in step 4 he moved his head towards left and found the image pin to be towards the left of the image and so he adjusted the image pin so that parallax was removed. Now another student Sumit placed a large thin vertical transparent glass slab between the object and the lens. Choose the correct statement(s) regarding the experiment

- (A) After step 3, distance between image pin and lens was more than distance between image and lens.
 (B) After step 3, distance between image pin and lens was less than distance between image and lens.

Just after slab is kept, if Amit moves his head towards right while looking at the image
 (C) from the end of the optical bench, he will see the image pin to be towards right of the actual image.

Just after slab is kept, if Amit moves his head towards right while looking at the image
 (D) from the end of the optical bench, he will see the image pin to be towards left of the actual image.

7. The transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition:

- (A) $4 \rightarrow 1$ (B) $3 \rightarrow 2$ (C) $4 \rightarrow 2$ (D) $5 \rightarrow 4$

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Paragraph for Questions 1 and 2

In a photoelectric effect, a point source of green light of power 40 W emits mono-energetic photons that can just emit photoelectrons from an isolated metallic sphere of radius $r = 1$ cm, placed at a distance of 1m from the light source. Now four other separate sources of violet, blue, red and yellow light emitting same number of photons as that of green light are brought near the source of green light.

Assume one photoelectron is emitted out of every 10^6 incident photons on sphere.

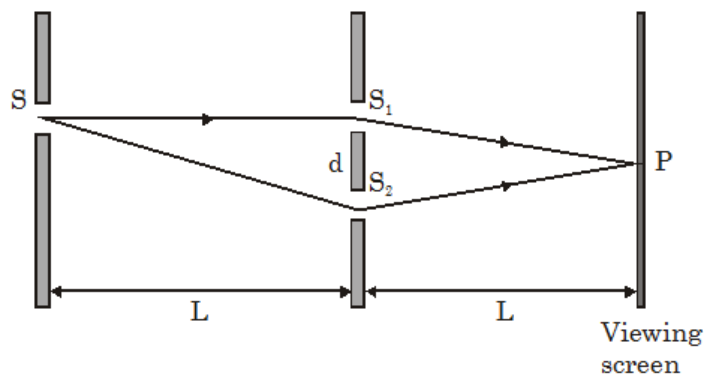
(Take $\lambda_{\text{violet}} = 4136 \text{ \AA}$, $\lambda_{\text{blue}} = 5000 \text{ \AA}$, $\lambda_{\text{red}} = 7200 \text{ \AA}$, $\lambda_{\text{green}} = 4963 \text{ \AA}$, $hc = 12408 \text{ eV \AA}$)

- No. of photoelectrons emitted from the sphere per second is
 (A) 2.5×10^9 (B) 2.5×10^8 (C) 5×10^9 (D) 7.5×10^9
- The potential of the sphere when emission of photoelectrons will stop is
 (A) 0.7 V (B) 0.5 V (C) 0.6 V (D) 0.8 V

Space for Rough Work

Paragraph for Questions 3 and 4

Monochromatic light of wavelength 625 nm passes through a very narrow slit S and then strikes a screen in which are two parallel slits, S_1 and S_2 as in figure. Slit S_1 is directly in line with S and at a distance of $L = 1.20$ m away from S, whereas S_2 is displaced a distance d to one side. The light is detected at point P on a second screen, equidistant from S_1 and S_2 . When either one of the slits S_1 and S_2 is open, equal light intensities are measured at point P. When both are open, the intensity is three times larger.



3. Find the minimum possible value for the slit separation d .

(A) 1.118 mm (B) 1.32 mm (C) 0.5 mm (D) 0.32 mm

4. If the slit separation is next higher value, what is the fringe width?

(A) $\frac{3\sqrt{3}}{10}$ mm (B) $\frac{3\sqrt{7}}{10}$ mm (C) $\frac{3\sqrt{5}}{10}$ mm (D) $\frac{3\sqrt{2}}{10}$ mm

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 21)

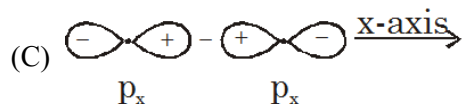
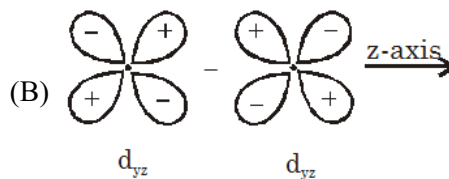
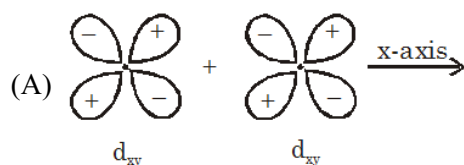
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Negative Marks : -1 In all other cases

1. Which of the following is correct IUPAC name for coordination isomer of



- (A) Diammine(oxalato)platinum(II)
 diammine(oxalato)paladate(II)
- (B) Diammine(oxalato)paladium(II)
 diammine(oxalato)platinate(II)
- (C) Tetraammineplatinum(IV)
 bis(oxalato)paladate(II)
- (D) Tetraamminepaladium(II)
 bis(oxalato)platinate(II)

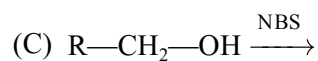
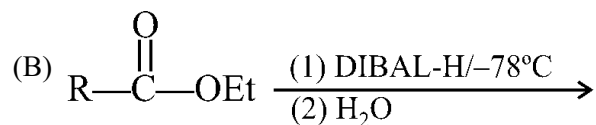
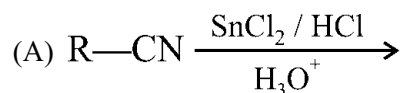
2. Which of the following overlap will form a molecular orbital having Gerade symmetry?



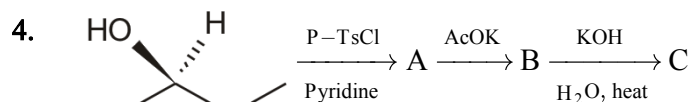
(D) Both (A) & (B)

Space for Rough Work

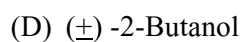
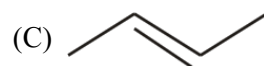
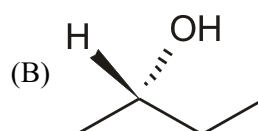
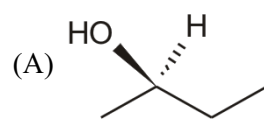
3. Which of the following reaction form aldehyde



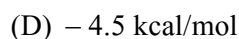
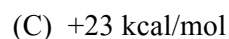
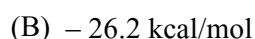
- (D) All of these



The major product (C) is



5. 1.6 gm $\text{CH}_4(\text{g})$ undergo combustion at constant volume at 300K and 2.5 kcal of heat is released. Enthalpy of combustion ($\Delta_c H$) of CH_4 will be:



Space for Rough Work

6. 2.08 kg of an equimolar mixture of ferrous oxalate (MW = 144) and ferric oxalate (MW = 376) require X moles of $K_2Cr_2O_7$ in acidic medium for complete oxidation. Then 'X' is -
(A) $16/3$ (B) 6 (C) $32/3$ (D) $10/3$
7. Insulin is dissolved in a suitable solvent & the osmotic pressure (π in atm) of solutions of various concentration 'C' in gm/cc is measured at $27^\circ C$. The slope of the plot π against 'C' is found to be 5×10^{-3} . The molecular weight of insulin is :
[R = 0.0821 litre-atm/mol-K]
(A) 4.9×10^5 (B) 4.9×10^3 (C) 4.9×10^6 (D) 4.9×10^4

Space for Rough Work

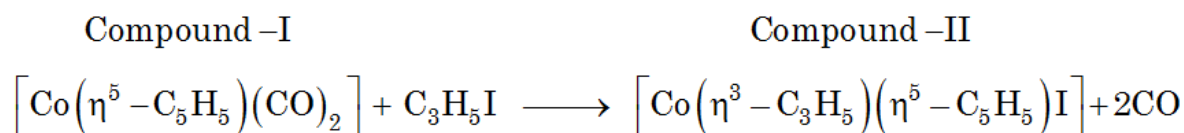
SEC-2 : (Maximum Marks: 28)

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1. When SiF_4 undergoes hydrolysis then which of the following statement is correct regarding to the mentioned hydrolysis :
 (A) H_2 is formed as one of the product
 (B) Reaction is redox
 (C) $\text{H}_2[\text{SiF}_6]$ is formed as one of the product
 (D) Partial hydrolysis

Space for Rough Work

2. Consider the following reaction



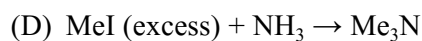
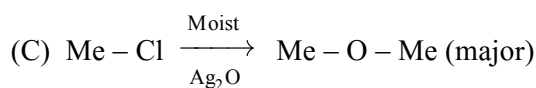
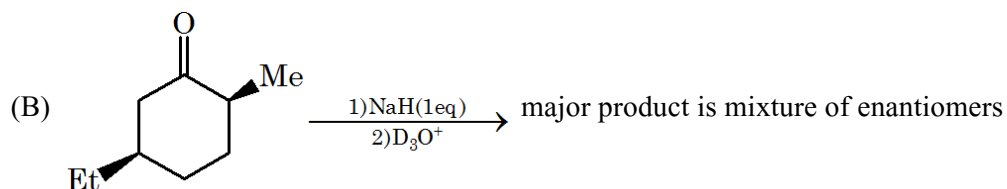
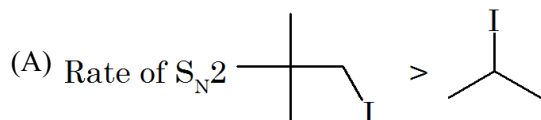
Which of the following is/are correct regarding the above reaction.

- (A) I and II have same value of EAN
- (B) CO formed in the reaction can be used to increase rate of reaction in Haber's process.
- (C) I and II both consist of trihapto ligands.
- (D) The above reaction is redox reaction
3. A metal (M) is extracted from its sulphide ore (X) by using following steps :
 After sedimentation ore is agitated by compressed air in the mixture of water and pineoil.
 Concentrated ore is burnt with excess of oxygen gives resulting product (p). (p) is either smelted with anthracite to produce metal or (p) is treated with dil. H_2SO_4 to dissolve (p) and filtered solution is electrolyzed for the deposition of metal.
 Which of the following steps is/are involved in the metallurgy of metal (M) from metal sulphide(X).
- (A) (X) $\rightarrow$ Froth Flotation $\rightarrow$ Roasting $\rightarrow$ Leaching $\rightarrow$ Carbon reduction $\rightarrow$ M
- (B) (X) $\rightarrow$ Froth Flotation $\rightarrow$ Roasting $\rightarrow$ Carbon reduction $\rightarrow$ M
- (C) (X) $\rightarrow$ Froth Flotation $\rightarrow$ Roasting $\rightarrow$ Leaching $\rightarrow$ Electrolytic reduction $\rightarrow$ M
- (D) (X) $\rightarrow$ Sedimentation $\rightarrow$ Froth Flotation $\rightarrow$ Roasting $\rightarrow$ Self reduction $\rightarrow$ M

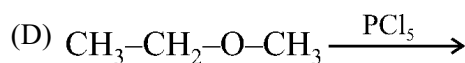
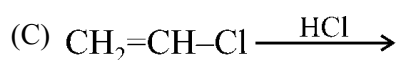
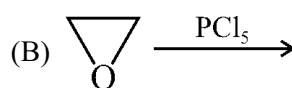
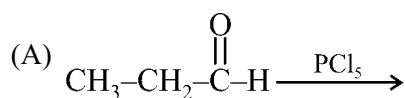
Space for Rough Work

4. Which of the following pair/s of ore consists the same metal?
- (A) Bauxite, Limonite (B) Haematite, Siderite
- (C) Cinnabar, Cassiterite (D) Galena, Cerrusite

5. Choose the incorrect option(s)



6. In which of the following reaction geminal dihalide formed as a major product :



Space for Rough Work

7. For a first order reaction carried out at constant P & T : $A(aq.) \rightarrow 2B(g) + 3C(g) + E(l)$.
If volume of gas B collected at $t = 1.386$ min is 200 ml and volume of gas C collected after sufficient time is 600 ml then
- (A) Rate constant of reaction is 0.5 min^{-1}
- (B) Volume of gas C collected after 2.772 min is 450 ml
- (C) Ratio of volume of gas C collected at $t = 1.386$ min and volume of gas B collected after sufficient time is $3/4$.
- (D) Half life time of reaction is 1.386 min.

Space for Rough Work

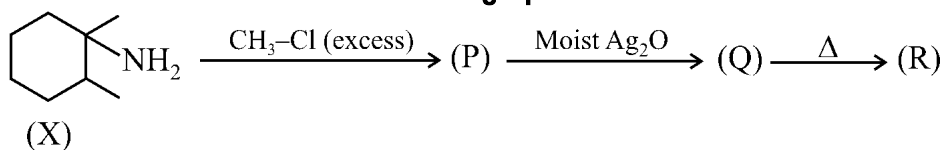
SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

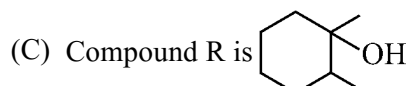
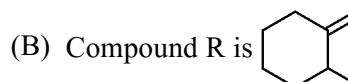
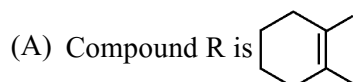
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2



1. Select correct statement



2. Total number of $\text{CH}_3\text{--Cl}$ molecule(s) consumed per molecule of compound (X) :

(A) 1

(B) 2

(C) 3

(D) 4

Space for Rough Work

Paragraph for Questions 3 and 4

One mole of a monoatomic ideal gas was isobarically heated from 300 K to 600 K at constant pressure of one atm. After that, it is subjected to reversible adiabatic expansion till the volume becomes $4\sqrt{2}$ of the initial value before isobaric process. After that it is subjected to isobaric cooling to original volume.

3. What is the net work done by the gas in the overall process ?

(A) $\frac{-450}{\sqrt{2}}R - 75R$

(B) $450R + \frac{75}{\sqrt{2}}R$

(C) $-450R + \frac{75}{\sqrt{2}}R$

(D) $\frac{450}{\sqrt{2}}R + 75R$

4. What is the change in enthalpy in adiabatic process ?

(A) $-750 R$

(B) $+500 R$

(C) $+750 R$

(D) $-500 R$

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Number of real solutions of equation $|x^2 - 2|x|| = 2^x$, is -
 (A) 1 (B) 2 (C) 3 (D) 4
2. The number of real solutions of the equation, $\cot^{-1}(x-1) + \cot^{-1}(6-x) = \cot^{-1}(x-2)$ is.
 (A) 0 (B) 1 (C) 2 (D) infinite

3. Let $g(x) = \lim_{n \rightarrow \infty} (2^{2n} (1 - f^n(x)))$

for $f: [0, 1] \rightarrow \mathbb{R}$, $f(x) = \sqrt{\frac{1+x}{2}}$

where $f^n(x) = \underbrace{f \circ f \circ \dots \circ f}_{n \text{ - times}}(x)$

Then which of the following is correct

- | | |
|---|---|
| (A) $g\left(\frac{1}{2}\right) = \frac{\pi^2}{18}$ | (B) $g\left(\frac{1}{2}\right) = \frac{-\pi^2}{18}$ |
| (C) $g\left(-\frac{1}{2}\right) = \frac{2\pi^2}{9}$ | (D) $g\left(-\frac{1}{2}\right) = \frac{4\pi^2}{9}$ |

Space for Rough Work

4. If $\left(\left(\left(e^{y-x} \right)^{y-2x} \right)^{y-3x} \right) \dots \text{upto } n \text{ terms} = 1$, then $\sum_{r=1}^{10} \frac{d^r y}{dx^r}$ is equal to -
- (A) 0 (B) 1 (C) $\frac{y}{x}$ (D) $\frac{x}{y}$
5. If $f(x) = \int 2^{\frac{\sin x}{\log_e 2}} \left(\frac{5 \cos x + \cos^3 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) dx$
and $f(0) = 1$ then value of $f(\pi/2)$ equals
- (A) $\sqrt{3}e$ (B) $\frac{\sqrt{3}e}{2}$ (C) $\frac{3e}{2}$ (D) $3e$
6. The least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is
- (A) $4\sqrt{3}r$ (B) $2\sqrt{3}r$ (C) $6\sqrt{3}r$ (D) $8\sqrt{3}r$
7. Find the area bounded by the curves $|y| + \frac{1}{2} \leq e^{-|x|}$.
- (A) $(2 - \ln 2)$ (B) $(2 - 2\ln 2)$ (C) $[3 - \ln 2]$ (D) $[3 - \ln 4]$

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A culture initially has 1 million number of bacteria. After two hours it is observed to be doubled. If the rate of growth is proportional to the number of bacteria present at time 't', then the number of bacteria after 7 hours is $10^6 (n)^{\frac{7}{n}}$, $n \in \mathbb{N}$. Which of the following is/are correct ?

(A) If $f(x) = (2x - 3\pi)^3 + 3x - \cos x$, then $\frac{d}{dx}(f^{-1}(x))$ at $x = \frac{9\pi}{2}$ is $\frac{1}{n}$

(B) Number of solutions of $|\cos x| = e^{-x}$ in $x \in (0, \pi)$ is $(n + 1)$

(C) If $f(x) = \sum_{\lambda=1}^m \left(x - \frac{2}{\lambda}\right) \left(x - \frac{1}{\lambda+1}\right)$ then $\lim_{m \rightarrow \infty} f(0) = n$

(D) Area bounded by the curves $y^2 = 12x$ and $x^2 = \frac{1}{4}y$ is n

Space for Rough Work

2. Which of the following can be solution of differential equation $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - x^2 = 0$ is (c_1 and c_2 are arbitrary constant) :-

(A) $y = e^x + c_1x + c_2$

(B) $y = \frac{x^3}{12} - \frac{c_1}{x} + c_2$

(C) $y = \frac{x^3}{12} + \frac{c_1}{x} + c_2$

(D) $y = \frac{x^3}{12} + e^x(c_1 + xc_2)$

3. Let $f(x)$ be a function satisfying $f(x) + f(x+2) = 20 \forall x \in \mathbb{R}$ then

(A) $f(x)$ is a periodic function

(B) $\int_{-3}^{13} f(x) dx = 80$

(C) $\int_{-1}^7 f(x) dx = 80$

(D) $f(x)$ is a many one function

Space for Rough Work

4.

If $f(x) = \int_0^1 f(xt) dt$, where $f(x)$ is a continuous function such that $f(1) = 2$ then

(A) $f(x)$ is a periodic function

(B) $f(4) = 2$

(C) $f(x)$ is an even function

(D) $f(4) = 5$

5.

Let $M = \int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta$, then

(A) $M > \frac{\pi}{3}$

(B) $M > \frac{\pi}{4}$

(C) $M < \frac{\pi^{\frac{3}{2}}}{3\sqrt{2}}$

(D) $M < \frac{2\pi}{3}$

Space for Rough Work

6. Let _____ for

$$n > 1, n \in \mathbb{I}, \lim_{n \rightarrow \infty} \int_0^{(n!+1)\frac{\pi}{2}} \left(\frac{\cos x + x \sin x}{x^2 + \cos^2 x} \right) dx = \ell \text{ and } \lim_{x \rightarrow \infty} (x^2 \ln(x \cot^{-1} x)) = m, \text{ then}$$

- (A) $\ell \in \mathbb{Q}, m \notin \mathbb{Q}$
 (B) $\ell \notin \mathbb{Q}, m \in \mathbb{Q}$
 (C) $[\ell] + [m] = 0$; ($[\cdot]$ denotes G.I.F.)
 (D) $\ell < 0 < m$
7. Let $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}, (a > 0)$ be the equation of curve then which of the following is/are correct
- (A) Equation of tangent at (x_1, y_1) on the curve is $\frac{x}{x_1^{1/3}} + \frac{y}{y_1^{1/3}} = a^{2/3}$ (x_1, y_1 are not equal to zero)
 (B) Length of portion of tangent intercepted between the coordinate axes is constant.
 (C) If tangent at any point on the curve meet the coordinate axes in A and B, then locus of midpoint of AB is $x^2 + y^2 = \frac{a^2}{4}$
 (D) Equation of normal at (x_1, y_1) on the curve is $xx_1^{1/3} - yy_1^{1/3} = x_1^{4/3} + y_1^{4/3}$

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A differentiable function $f(x)$ satisfying the following integral equation $f(x) = (\log_e x)^2 - \int_1^e \frac{f(t)}{t} dt$.

Suppose $f(x) + \frac{1}{6} = g(x)$, then

1. If $\frac{dz}{dx} - \frac{1}{x \ln x} \cdot z = g(x)$, then z is given by-
(where 'C' is integration constant)

(A) $x \ln^2 x + (C - x) \ln x$	(B) $(x + C) \ln x$
(C) $x(\ln x - 1) + C \ln x$	(D) None of these
2. If the area of the figure bounded by the tangent line of $y = f(x)$ at the point $(e, f(e))$, the curve $y = f(x)$ and the line $x = 1$ is A, then the smallest integer greater than or equal to 'A' is-

(A) 0	(B) 1	(C) 2	(D) 3
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Space for Rough Work

Paragraph for Questions 3 and 4

$$\text{Let } f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{|x+1|^n + x^2}{|x| + x^{2n}} & -6 \leq x < 0 \\ \{\sin x\} & 0 \leq x < 6 \end{cases}, \text{ where } \{.\} \text{ denotes fractional part function.}$$

on the basis of above information, answer the following questions :

3. Number of points at which $f(x)$ is not differentiable in $[-6, 6]$ -
 (A) 6 (B) 5 (C) 4 (D) 3
4. Number of solutions of the equation $f(x) = \frac{x}{6}$ in $[-6, 6]$
 (A) 2 (B) 3 (C) 4 (D) 5

Space for Rough Work

Space for Rough Work